

FlowState calibration & validation report

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Client summary

Baseline gate FAILED (docs/FRISCO_PROTOCOL.md section 6): C1 link flows (calibration), C3 speeds (calibration), C1 link flows (validation), C3 speeds (validation), C4 wave speed (calibration). No strategy recommendation may be made from this model.

- replicates PASS: Replicates: 20 seeded replicate(s); the protocol scores every check over at least 20. Source: [FlowState] CLAUDE.md section 0.6; protocol section 4.
- days PASS: Day sets: the calibration-day artifact holds exactly the split's 5 calibration day(s) and the validation-day artifact its 4 validation day(s), none shared. Source: [FlowState] protocol section 3.2.
- quality PASS: Targets quality-masked: calibration days: runs/p1_rehearsal/dq/data_quality.json (sha256 8cae907dec40): 24 masked detector-day(s) (24 excluded whole), 912 reading(s) set aside; validation days: runs/p1_rehearsal/dq/data_quality.json (sha256 8cae907dec40): 19 masked detector-day(s) (19 excluded whole), 720 reading(s) set aside. Source: [FlowState] protocol section 2.2.
- C1 FAIL: Link flows, calibration days: GEH < 5 on 36.1 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 36.1 %, 95 % interval 34.3 % to 37.9 %); the target is >= 85.0 %, short by 48.9 percentage points; hours anchored at the study period's start (1800 s). Source: [federal] FHWA TAT Vol. III 2004 (profile fhwa_default).
- C2 FAIL (reported, not part of the gate): Link flows, Texas criterion, calibration days: GEH < 3 on 25.2 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 36.1 %, 95 % interval 34.3 % to 37.9 %); the target is >= 100.0 %, short by 74.8 percentage points; hours anchored at the study period's start (1800 s). Source: [federal] TxDOT TSAP ch. 13 (profile txdot_tsap_ch13); reported, not gating.
- C3 FAIL: Speeds, calibration days: RMSPE of 15-minute station mean speeds 38.3 % (mean over 20 replicate(s), 95 % interval 37.5 % to 39.2 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 23.3 percentage points (diagnostics: 5 min 39.8 %; 60 min 34.8 %). Source: common microsimulation practice (CLAUDE.md section 7.1), cited in the report.
- C6 PASS: Bottlenecks, calibration days: 0 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold. Source: [FlowState] protocol section 5.
- C1 FAIL: Link flows, validation days: GEH < 5 on 33.0 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 33.0 %, 95 % interval 30.9 % to 35.1 %); the target is >= 85.0 %, short by 52.0 percentage points; hours anchored at the study period's start (1800 s). Source: [federal] FHWA TAT Vol. III 2004 (profile fhwa_default).
- C2 FAIL (reported, not part of the gate): Link flows, Texas criterion, validation days: GEH < 3 on 23.2 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 33.0 %, 95 % interval 30.9 % to 35.1 %); the target is >= 100.0 %, short by 76.8 percentage points; hours anchored at the study period's start (1800 s). Source: [federal] TxDOT TSAP ch. 13 (profile txdot_tsap_ch13); reported, not gating.
- C3 FAIL: Speeds, validation days: RMSPE of 15-minute station mean speeds 37.9 % (mean over 20 replicate(s), 95 % interval 37.0 % to 38.7 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 22.9 percentage points (diagnostics: 5 min 39.7 %; 60 min 33.2 %). Source: common microsimulation practice (CLAUDE.md section 7.1), cited in the report.
- C6 PASS: Bottlenecks, validation days: 1 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold. Source: [FlowState] protocol section 5.
- C4 FAIL: Wave speed: a backward front in 20 of 20 replicate(s) (100.0 %; at least 80.0 % needed, a replicate without a front counting as a miss); simulated backward wave speed 5.7 km/h, 95 %

interval 5.5 to 5.8 km/h over those replicates (stack detector); the band is 14-22 km/h, outside it by 8.3 km/h; observed on the calibration days (detector cross-correlation): median 19.1 km/h from 7 of 13 station pairs. Source: empirical stop-and-go literature (flowstate_core.constants.WAVE_SPEED_BAND_KMH).

- C5 PASS: Collisions: none in 20 run(s). Source: [FlowState] CLAUDE.md section 3.3; owner decision 2026-10-04.

What we are confident about, and what we are not

Statement	Confident	Basis
Traffic counts at the detectors (link flows, C1)	no	Link flows, calibration days: GEH < 5 on 36.1 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 36.1 %, 95 % interval 34.3 % to 37.9 %); the target is ≥ 85.0 %, short by 48.9 percentage points; hours anchored at the study period's start (1800 s). Link flows, validation days: GEH < 5 on 33.0 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 33.0 %, 95 % interval 30.9 % to 35.1 %); the target is ≥ 85.0 %, short by 52.0 percentage points; hours anchored at the study period's start (1800 s).
Speeds at the detectors, 15-minute averages (C3)	no	Speeds, calibration days: RMSPE of 15-minute station mean speeds 38.3 % (mean over 20 replicate(s), 95 % interval 37.5 % to 39.2 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 23.3 percentage points (diagnostics: 5 min 39.8 %; 60 min 34.8 %). Speeds, validation days: RMSPE of 15-minute station mean speeds 37.9 % (mean over 20 replicate(s), 95 % interval 37.0 % to 38.7 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 22.9 percentage points (diagnostics: 5 min 39.7 %; 60 min 33.2 %).
Where and when the slowdowns form (bottlenecks, C6)	yes	Bottlenecks, calibration days: 0 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold. Bottlenecks, validation days: 1 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold.
Stop-and-go wave speed (C4)	no	Wave speed: a backward front in 20 of 20 replicate(s) (100.0 %; at least 80.0 % needed, a replicate without a front counting as a miss); simulated backward wave speed 5.7 km/h, 95 % interval 5.5 to 5.8 km/h over those replicates (stack detector); the band is 14-22 km/h, outside it by 8.3 km/h; observed on the calibration days (detector cross-correlation): median 19.1 km/h from 7 of 13 station pairs.
No simulated collisions (C5)	yes	Collisions: none in 20 run(s).
No permanent standstill (gridlock) in any run	yes	no run of 20 had a queue standing with no discharge for 10 min or more
Fuel (model estimate, SUMO HBEFA emission class; not validated against measured fuel)	no	fuel is not validated: no measured fuel was compared, so every fuel figure is a model estimate (docs/FRISCO_PROTOCOL.md section 8.6)
Effects of the strategies	no	not delivered: the model did not reproduce the corridor
Robustness of strategy effects to driver and demand uncertainty	no	not evaluated in this report (docs/FRISCO_PROTOCOL.md section 8.5)

Statement	Confident	Basis
Waiting time on ramps and before entering the network	yes	counted: every run records total delay and travel time including waiting (docs/FRISCO_PROTOCOL.md section 8.2); no strategy recommendation is made because the run set has no single baseline (do-nothing) group to compare against

Strategy results

This report contains no strategy recommendations: the model did not reproduce the corridor (the baseline gate failed), so strategy results are not delivered as findings.

Data, days and assumptions

- Calibration days: 2026-09-02, 2026-09-03, 2026-09-08, 2026-09-15, 2026-09-16. Validation days: 2026-09-01, 2026-09-09, 2026-09-10, 2026-09-17. Drawn by the protocol's seeded split (seed 20261004, docs/FRISCO_PROTOCOL.md section 3).
- Excluded detector 3240: S792 lane-3 loop chatters: 14 percent occupancy against 27-32 percent on its neighbours, 78 percent null night samples, count rises 775 veh/h into a bracket nothing enters (docs/ONBOARDING_MNDOT.md section 7 item 2, 2026-09-24)
- Observed data: artifacts/p1_rehearsal_2026-10-04/observations_calibration.json, dates 20260902, 20260903, 20260908, 20260915, 20260916; mean over dates per window (weekday typical profile).
- Ramp volumes estimated from mainline differences, and every split assumption behind them, are listed in the study's ramp-estimation artifact, not in this run set (docs/FRISCO_PROTOCOL.md section 2.3).
- Strategy assumptions: each strategy arm's penetration, compliance and settings are as its label states; they are assumptions, not measured behaviour.

Limits of this study

- Single corridor: the results describe this corridor, period and day set; transfer to other corridors is not established.
- Model-form uncertainty: the car-following and lane-change models' own assumptions are not captured by the seed-to-seed intervals.
- Every fuel figure is a model estimate (SUMO HBEFA emission class), not validated against measured fuel; differences in fuel between configurations are model predictions.
- Compliance and strategy assumptions: strategy results hold only for the settings simulated, under the configured driver population and demand.
- Results are reported as they came out, including failures.

Provenance

Profile: fhwa_tat3_2004. Seeds: 134183728835869882, 165503670820534583, 2378473973028931053, 3011106312394044631, 3747978530954135749, 3944094060050347669, 4910985839736976611, 5690692725577505498, 6134032994440706937, 6143473282319009404, 6538422657834023852, 661281422688282993, 677105600768189526, 6904272788004776631, 6914975401685141156, 6953598295321596746, 7382187975121682178, 8026499204807041784, 8557154790156791364, 887972120279483394.

Measurement window: each run's configured warm-up is discarded from every metric (warm-up per run, in seconds: 1800). Travel times keep whole journeys that begin inside the window and are measured over [1089.1, 11071.8] m. Fuel per vehicle-km, a model estimate (SUMO HBEFA emission class; not validated against measured fuel), remains a whole-run ratio unless the run records a post-warm-up fuel total.

Insertion: 647080 vehicles planned over 20 run(s), 636724 departed (0.984 of plan on average, lowest 0.982), 30317.0 arrived per run (over 20 of 20); verdict: ok.

Weave exits at on-ramp 745524613 (exit C-D split 18208090): 599 of 30717 reached exiters (1.95 %) were given up at the gore's end and rerouted through over 20 run(s), within the 2 % threshold; the exit's link flow is short by that count and every mainline link downstream carries it.

Weave exits at on-ramp 769818012 (exit off-ramp 18207598): 990 of 77870 reached exiters (1.27 %) were given up at the gore's end and rerouted through over 20 run(s), within the 2 % threshold; the exit's link flow is short by that count and every mainline link downstream carries it.

Run	Config hash	Seed	Tier	seeded	Wall time [s]
134183728835869882	395a111cb991	134183728835869882	micro	seeded=False	1141
165503670820534583	395a111cb991	165503670820534583	micro	seeded=False	1098
2378473973028931053	395a111cb991	2378473973028931053	micro	seeded=False	1057
3011106312394044631	395a111cb991	3011106312394044631	micro	seeded=False	1117
3747978530954135749	395a111cb991	3747978530954135749	micro	seeded=False	1540
3944094060050347669	395a111cb991	3944094060050347669	micro	seeded=False	1121
4910985839736976611	395a111cb991	4910985839736976611	micro	seeded=False	1151
5690692725577505498	395a111cb991	5690692725577505498	micro	seeded=False	1000
6134032994440706937	395a111cb991	6134032994440706937	micro	seeded=False	1244
6143473282319009404	395a111cb991	6143473282319009404	micro	seeded=False	1283
6538422657834023852	395a111cb991	6538422657834023852	micro	seeded=False	1598
661281422688282993	395a111cb991	661281422688282993	micro	seeded=False	1324
677105600768189526	395a111cb991	677105600768189526	micro	seeded=False	1585
6904272788004776631	395a111cb991	6904272788004776631	micro	seeded=False	1451
6914975401685141156	395a111cb991	6914975401685141156	micro	seeded=False	1445
6953598295321596746	395a111cb991	6953598295321596746	micro	seeded=False	1151
7382187975121682178	395a111cb991	7382187975121682178	micro	seeded=False	951.8
8026499204807041784	395a111cb991	8026499204807041784	micro	seeded=False	1348
8557154790156791364	395a111cb991	8557154790156791364	micro	seeded=False	1496
887972120279483394	395a111cb991	887972120279483394	micro	seeded=False	1149

Package versions (from run metadata)

- eclipse-sumo: 1.27.1
- flowstate_core: 2.0.0
- libsumo: 1.27.1
- microsim: 2.0.0
- numpy: 2.5.2
- pandas: 3.0.5
- pyarrow: 25.0.1
- python: 3.12.15

Calibration artifacts used

Artifact	data_hash
artifacts/idm_i24_capacity_amax_k1.0.json	aa97dd93d2bf250ea23c3624f91afe8d7035a672fd13efbd53013709e12c7d56

Observed data

The link-flow and segment-speed criteria are scored against this observed artifact: hourly volumes formed from the fully observed windows of each hour at every mainline station, and mean speeds per station segment (each station owns the span to the midpoints with its neighbours). Simulation time zero is the artifact's local start time; the run's warm-up, and any window the run does not cover to its end, are excluded, and a station-window the detector did not measure is skipped, never imputed — the

coverage rows say how much of the grid was compared. A station whose cross-section lies outside the simulated position span is excluded from both comparisons rather than scored against a simulated flow of zero; the rows say how many were. When the artifact carries a detector-estimated backward wave speed, it is printed here as context for the band the simulated wave speed is scored against — it is a property of the corridor, not a target, and no criterion is evaluated from it.

Field	Value
artifact	artifacts/pl_rehearsal_2026-10-04/observations_calibration.json
corridor	I-94 WB
source provider	MnDOT RTMC Mayfly API
source dates	20260902, 20260903, 20260908, 20260915, 20260916
source url	https://data.dot.state.mn.us/mayfly
aggregation	mean over dates per window (weekday typical profile)
local clock time of simulation t = 0	05:30
observation window [s]	300
mainline stations compared	14
observation windows	48
windows inside the measurement window	42
station-windows with an observed flow [fraction]	1
station-windows with an observed speed [fraction]	1
link-hour comparisons (pooled over replicates)	840
speed cells compared (pooled over replicates)	11760
replicates scored against the observations	20
stations excluded (outside the simulated span)	0
detector-estimated backward wave speed (context, not a criterion)	median 19.1 km/h (IQR 18–23.9) from 7 of 13 station pairs; leave-one-date-out 18.5–20.8 km/h over 5 dates (fewest 5 pairs); the model's band is 14–22 km/h

Model integrity

What the runs' metadata records about the simulation itself misbehaving (docs/CONTRACTS.md). A collision in a car-following simulation is a model defect, not a traffic outcome, and so is a lock: vehicles standing with no discharge past a point while a queue builds behind it, read from each run's space-time table and vehicle table (validation.locks). A run whose metadata carries no counter, or that has neither table, is reported as not recorded, never as zero.

- Collisions: 0 over 20 run(s) that record the counter; per run 0 [0, 0] (two-sided 95 % t-interval, n = 20); 0 per 1,000 departed vehicles (0 over 636724 departed in 20 run(s), whole runs including warm-up).
- Locks: 0 of 20 run(s) locked (0 %; 95 % Clopper-Pearson interval 0–16.8 %). A lock is a queue standing with no discharge past a point for at least 10 min.
- Weave rules at on-ramp 745524613 (exit C-D split 18208090): W1b released 257 of 20340 entrance departures (1.26 %) over 20 run(s), above the 1 % design bound; W2 (20 of 20 run(s) with all three guards on): handback skips 3546, close-leader withholds 427, opposing deferrals 6985, of them vetoes 6908.
- Weave rules at on-ramp 769818012 (exit off-ramp 18207598): W1b released 1 of 91644 entrance departures (0.00109 %) over 20 run(s), within the 1 % design bound; W2 (20 of 20 run(s) with all three guards on): handback skips 2880, close-leader withholds 101, opposing deferrals 16767, of them vetoes 16011.
- Forced lane changes at on-ramp 18207436 (scripted merge): 1527 of 15318 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.

- Forced lane changes at on-ramp 178547099 (scripted merge): 2980 of 19808 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.
- Forced lane changes at on-ramp 745524613 (weave section): 9924 of 25057 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.
- Forced lane changes at on-ramp 769818012 (weave section): 9265 of 92735 completed change(s) over 20 run(s) were made under SUMO's forced lane-change mode, in which SUMO refuses a change only on an overlap — the path around its own lane-change safety checks.

Acceptance criteria

Criterion	Value	Threshold	Evaluated	Result
link_flows_geh	0.4607	GEH < 5 for > 85% of link-hour comparisons	yes	FAIL – fraction of comparisons with GEH < 5
wave_speed	5.669	backward wave speed in [14, 22] km/h (emergent, unseeded)	yes	FAIL – detector: stack: slant-stack peak of the two-way demeaned field on 15 s x 75 m bins over front speeds [-40, -2] km/h in 0.25 km/h steps, peak/median contrast >= 3, edge peaks rejected
ring_emergence	–	Sugiyama ring emergence benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
ring_dampening	–	Stern single-AV dampening benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
n_seeds	20	n_seeds >= 20	yes	PASS
sensitivity_grid	–	penetration {1%, 2%, 5%, 10%, 15%, 20%} x compliance {25%, 50%, 80%, 100%} grid published with CIs	no	NOT EVALUATED – not evaluated: input not supplied
no_collisions	0	zero SUMO collisions in every run, each run recording the counter	yes	PASS – no collision in 20 run(s); FlowState internal standard (CLAUDE.md §3.3; owner decision 2026-10-04), not an FHWA or DOT criterion
no_locks	0	no run locked (no queue standing without discharge for 10 min or more), every run recorded	yes	PASS – no lock in 20 run(s); FlowState internal standard (CLAUDE.md §0.1; docs/I94_COLLAPSE_DIAGNOSIS.md §7, 2026-10-07), not an FHWA or DOT criterion
wlb_release_share (on-ramp 745524613)	0.01264	reported beside no_locks, not gating: W1b releases as a share of the entrance's departures, pooled over the runs; above 1% flagged in Limitations	yes	REPORTED – reported, not gating: 257 of 20340 entrance departures released into the paired exit by W1b over 20 run(s) (1.26 %), ABOVE the 1% design bound (flagged in Limitations); W2 over the runs it was on in: handback skips 3546, close-leader withholds 427, opposing deferrals 6985, of them vetoes 6908; Amendment 4 (docs/FRISCO_PROTOCOL.md, 2026-10-07): weave rules W1b and W2 are part of the model; a FlowState disclosure, not an FHWA or DOT criterion

Criterion	Value	Threshold	Evaluated	Result
wlb_release_share (on-ramp 769818012)	1.091e-05	reported beside no_locks, not gating: W1b releases as a share of the entrance's departures, pooled over the runs; above 1% flagged in Limitations	yes	REPORTED – reported, not gating: 1 of 91644 entrance departures released into the paired exit by W1b over 20 run(s) (0.00 %), within the 1% design bound; W2 over the runs it was on in: handback skips 2880, close-leader withholds 101, opposing deferrals 16767, of them vetoes 16011; Amendment 4 (docs/FRISCO_PROTOCOL.md, 2026-10-07): weave rules W1b and W2 are part of the model; a FlowState disclosure, not an FHWA or DOT criterion

A row marked not evaluated is never a pass: either no input was supplied, or the value was measured with a recipe this profile does not accept (the row's result says which). An unevaluated criterion counts as failing. A row marked not recorded is never a pass either: some run in the set carries no record of what the row checks, and a missing record is not a zero. The no_collisions row is a FlowState model-integrity requirement, not an FHWA criterion: every run must record its SUMO collision counter, and the total must be zero. So is the no_locks row: every run must carry a lock record, and no run may lock.

Wave-speed criterion input: mean over 20 of 20 unseeded replicate(s) of group merge scripted on on-ramp 18207436 + merge scripted on on-ramp 178547099 + merge weave on on-ramp 745524613 + merge weave on on-ramp 769818012 (395a111cb991), measured with the stack detector on its own bins. A row marked REPORTED is a disclosure, never a pass or a fail: Amendment 4 (docs/FRISCO_PROTOCOL.md, 2026-10-07) puts weave rules W1b and W2 into the model at every weaving section, and each wlb_release_share row states the share of the entrance's departures W1b sent into the paired exit after standing its dwell at the end of the auxiliary lane, pooled over every run of the set, with W2's counters. A share above 1 % is listed under Limitations.

Speed criterion by time aggregation

The segment-speed criterion compares a replicate mean with one recorded day. The floor column is the recorded field against its own three-window moving average: below that resolution the day does not repeat itself, so no ensemble mean can score under the floor. The criterion row keeps its native resolution; this table says how much of its value is resolution.

Aggregation	RMSPE, replicate mean vs observed	Floor: observed vs its own three-window average
5 min (criterion)	0.5445	0.03733
15 min	0.5429	0.07091
30 min	0.5388	
60 min	0.5203	
whole period	0.4944	

Metrics

Mean with two-sided t-distribution confidence bounds at the 95 percent level over n replicates. Rows flagged underpowered have fewer than 20 replicates and must not be quoted as headline results. Travel-time rows rest on the vehicles counted by the n_travel_time_veh row, which are those that entered within the measurement window and completed the span.

The wave_speed_kmh row is the metrics detector's diagnostic reading (standard); the acceptance criterion above is measured separately with the profile's stack detector on its own bins, so the two values can differ.

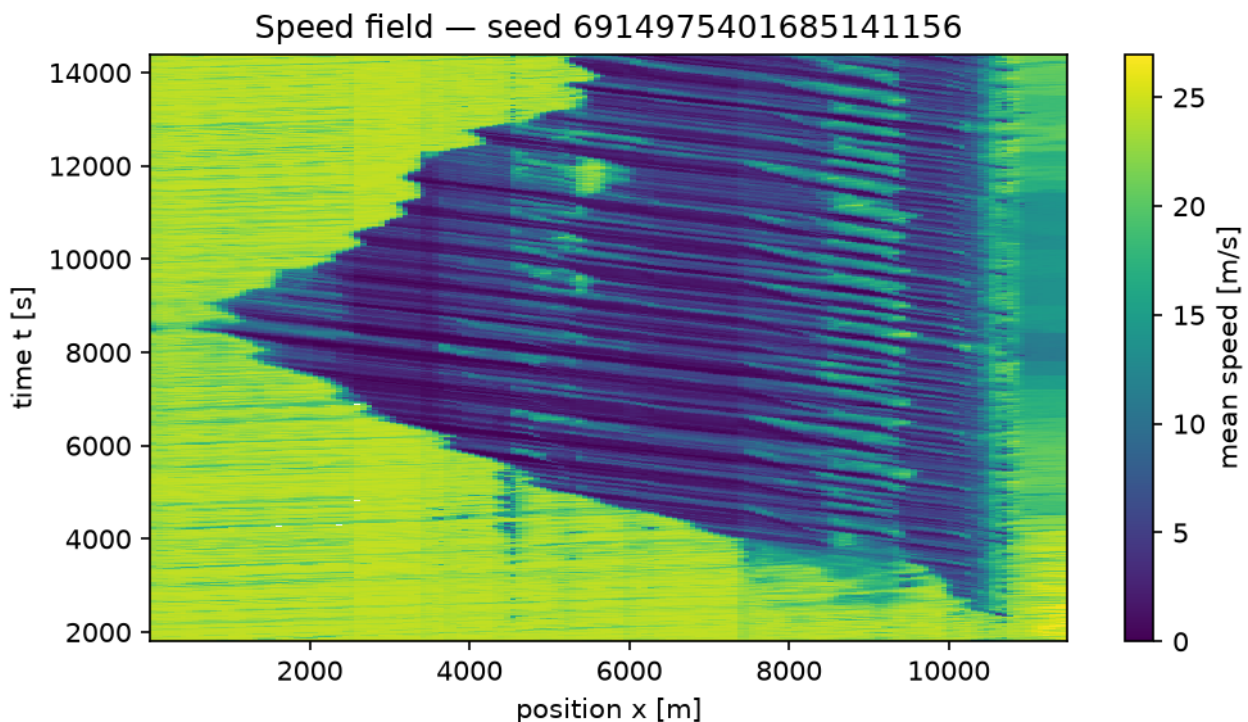
Rows marked (model estimate) carry fuel (model estimate, SUMO HBEFA emission class; not validated against measured fuel). SUMO computes fuel from its HBEFA emission classes; this study has measured no fuel to check it against (docs/FRISCO_PROTOCOL.md section 8.6).

merge scripted on on-ramp 18207436 + merge scripted on on-ramp 178547099 + merge weave on on-ramp 745524613 + merge weave on on-ramp 769818012 (395a111cb991)

Replicates: n = 20 distinct seeds (134183728835869882, 165503670820534583, 2378473973028931053, 3011106312394044631, 3747978530954135749, 3944094060050347669, 4910985839736976611, 5690692725577505498, 6134032994440706937, 6143473282319009404, 6538422657834023852, 661281422688282993, 677105600768189526, 6904272788004776631, 6914975401685141156, 6953598295321596746, 7382187975121682178, 8026499204807041784, 8557154790156791364, 887972120279483394); replicate criterion (n_seeds >= 20): PASS.

Not delivered: the model did not reproduce the corridor (the baseline gate failed, docs/FRISCO_PROTOCOL.md section 6). Strategy runs are kept for internal learning only; their numbers are not findings and are not shown.

Speed contours



Space-time mean-speed contour, seed 6914975401685141156

Limitations

- At the on-ramp 745524613 weave, W1b sent 257 of 20340 entering vehicles (1.26 %, over 20 run(s)) into the paired exit after each stood the W1b dwell at the end of the auxiliary lane — above the 1 % design bound of Amendment 4. W1b is a safeguard that keeps the simulation running, not measured driver behaviour: a release ends either a lock at the gore or a long ordinary wait, the share counts interventions, not realism, and the exit's and the mainline's flows carry those vehicles.
- Results cover a single corridor/scenario family; transfer to other corridors is not established.
- Model-form uncertainty (IDM/SUMO car-following assumptions, effective single-pipe demand representation) is not captured by seed-to-seed confidence intervals.
- AV penetration and compliance are swept assumptions, not measured behavior; conclusions hold

only within the swept grid.

- Metrics from fewer than 20 replicates are flagged underpowered and are diagnostics, not claims.
- Every fuel figure is a model estimate (SUMO HBEFA emission class), not validated against measured fuel; differences in fuel between configurations are model predictions.
- Measurement window: each run's configured warm-up is discarded from every metric (warm-up per run, in seconds: 1800). Travel times keep whole journeys that begin inside the window and are measured over [1089.1, 11071.8] m. Fuel per vehicle-km, a model estimate (SUMO HBEFA emission class; not validated against measured fuel), remains a whole-run ratio unless the run records a post-warm-up fuel total.
- The observed comparison covers the artifact's stations, windows and days only; station-windows the detectors did not measure are excluded from both criteria rather than filled in, so the coverage rows bound how much of the corridor and period the two rows actually speak for.